Faculty of Computer Science and Mathematics

Universiti Malaysia Terengganu

LAB REPORT

CSE3023-Web-Based Application Development

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Semester	2025/26-2
Lab Report	2
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TASK1: Data Sharing in Servlet

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Login Page</title>
</head>
<body>
  <h2>Login</h2>
  <form action="LoginServlet" method="POST">
    <label for="username">Username:</label>
    <input type="text" id="username" name="username"><br><br>

    <label for="password">Password:</label>
    <input type="password" id="password" name="password"><br><br>
    <input type="submit" value="Login">
  </form>
</body>
</html>
```

```
package SourcePackages;

import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.RequestDispatcher;
```

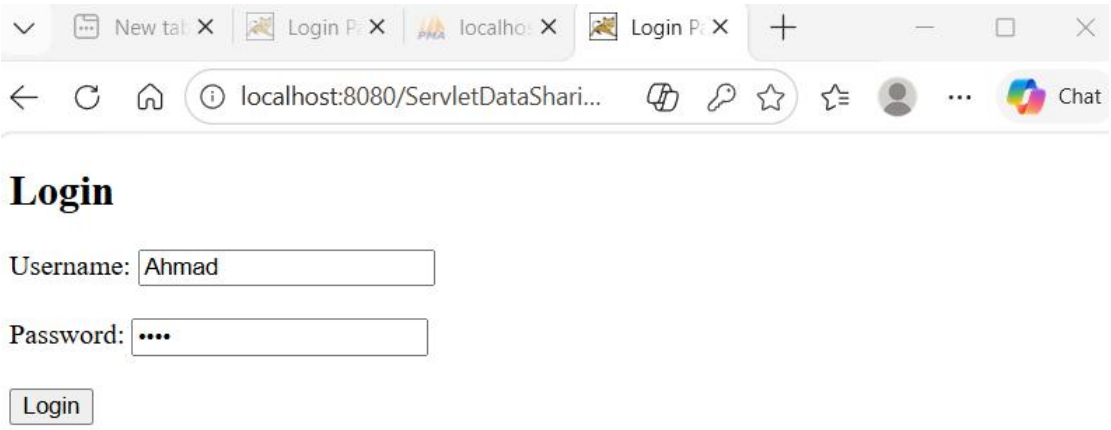
```
public class LoginServlet extends HttpServlet {

    protected void processRequest(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            /* TODO output your page here. You may use following sample code. */
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Servlet LoginServlet</title>");
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>Servlet LoginServlet at " + request.getContextPath() +
"</h1>");
            out.println("</body>");
            out.println("</html>");
        }
    }


    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }


    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {


    String user = request.getParameter("username");
    String pass = request.getParameter("password");
```

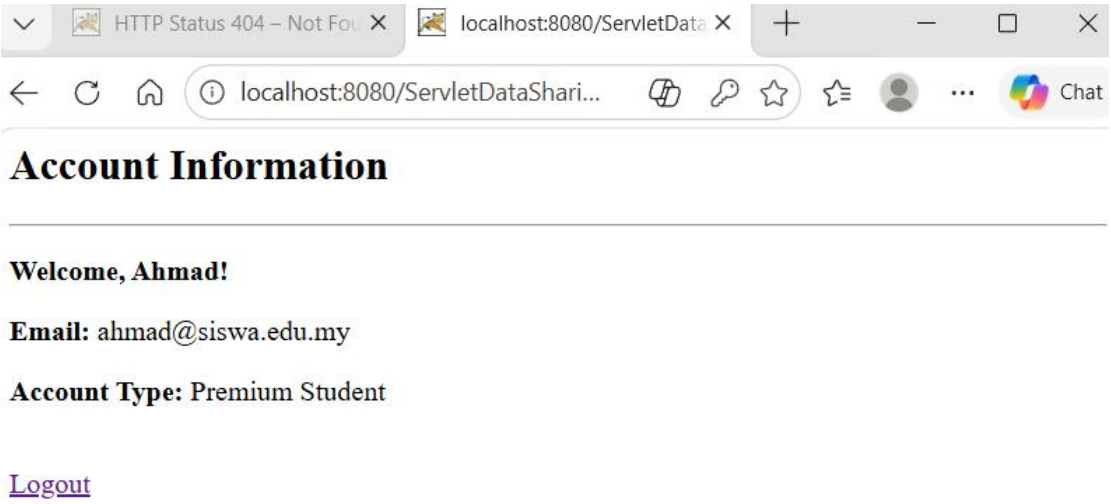
```
    // Semakan kata laluan (Authentication)
    if (user.equals("Ahmad") && pass.equals("4567")) {


    // Data Sharing: Menyimpan maklumat tambahan ke dalam 'request'
    request.setAttribute("accountType", "Premium Student");
    request.setAttribute("email", "ahmad@siswa.edu.my");

    // Request Forwarding ke AccountServlet
    RequestDispatcher rd = request.getRequestDispatcher
    ("AccountServlet");
    rd.forward(request, response);

  } else if (user.equals("Siti") && pass.equals("1234")) {
     // Contoh untuk pengguna lain
     request.setAttribute("accountType", "Standard Student");
     request.setAttribute("email", "siti@siswa.edu.my");
     RequestDispatcher rd = request.getRequestDispatcher
     ("AccountServlet");
     rd.forward(request, response);

} else {
    // Jika login gagal
    response.setContentType("text/html");
    PrintWriter out = response.getWriter();
    out.println("<h3 style='color:red'>Login Failed! Invalid credentials.</h3>");
    out.println("<a href='login.html'>Try Again</a>");
  }
}




    @Override
    public String getServletInfo() {
        return "Short description";
```

```
    }

}



//Servlet AccountServle
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class AccountServlet extends HttpServlet {
        protected void doPost(HttpServletRequest request, HttpServletResponse
        response) throws ServletException, IOException {

            response.setContentType("text/html");
            PrintWriter out = response.getWriter();

            // Mengambil semula parameter dari borang HTML asal
            String username = request.getParameter("username");

            // Mengambil data yang dikongsi oleh LoginServlet

            String accountType = (String) request.getAttribute("accountType");
            String email = (String) request.getAttribute("email");

            out.println("<html><body>");
            out.println("<h2>Account Information</h2>");
            out.println("<hr>");
            out.println("<p><strong>Welcome, " + username + "!</strong></p>");
            out.println("<p><b>Email:</b> " + email + "</p>");
            out.println("<p><b>Account Type:</b> " + accountType + "</p>");
            out.println("<br>");
            out.println("<a href='login.html'>Logout</a>");
```

```
        out.println("</body></html>");
    }
}
```

OUTPUT:





Reflections

1. What have you learned from this exercise?

- This exercise teaches the implementation of dynamic web applications by managing data sharing and request forwarding between Servlets. I have learned to use request.setAttribute() and getAttribute() to pass data such as account types or emails across components while using the RequestDispatcher to navigate from a LoginServlet to an AccountServlet.

2. What are the common methods used in Java Servlet?

- Common methods used in Java Servlets are doPost() for processing sensitive form data like login credentials and doGet() to retrieve or display data. Other methods are processRequest() for centralizing request logic, ini() for resource initialization such as instantiating a DAO and setAttribute()/ getAttribute() for sharing objects within the request scope.

TASK4: Using Servlets for Database CRUD Operations

CODE:

```
<!DOCTYPE html>
<html>
    <head>
        <title>User Registration</title>
        <meta charset="UTF-8">
    </head>
    <body>
        <h2>Add New User</h2>
        <form action="InsertServlet" method="POST">
            <label>Username:</label><br>
            <input type="text" name="username" required><br><br>

            <label>Password:</label><br>
            <input type="password" name="password" required><br><br>
```

```
            <label>Roles:</label><br>
            <input type="text" name="roles" required><br><br>

            <input type="submit" value="Submit">
        </form>
    </body>
</html>
```

```
package com.lab.controller;

import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import javax.servlet.annotation.WebServlet;


@WebServlet(name = "InsertServlet", urlPatterns = {"/InsertServlet"})
public class InsertServlet extends HttpServlet {

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        String username = request.getParameter("username");
        String password = request.getParameter("password");
        String roles = request.getParameter("roles");

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
```

```
            Connection conn =
DriverManager.getConnection("jdbc:mysql://localhost:3306/cse3023", "root",
"admin");

            String sql = "INSERT INTO users (username, password, roles) VALUES
(?, ?, ?)";
            PreparedStatement pstmt = conn.prepareStatement(sql);
            pstmt.setString(1, username);
            pstmt.setString(2, password);
            pstmt.setString(3, roles);
            pstmt.executeUpdate();

            pstmt.close();
            conn.close();

            response.sendRedirect("ViewServlet");

        }catch (Exception e) {
    response.setContentType("text/html");
    response.getWriter().println("SQL Error: " + e.getMessage());
    e.printStackTrace();
}
        }
}
```

```
package com.lab.controller;

import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
```

```
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/**
 *
 * @author Ikhwan
 */
@WebServlet(name = "ViewServlet", urlPatterns = {"/ViewServlet"})
public class ViewServlet extends HttpServlet {


    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        out.println("<h2>User List</h2>");
        out.println("<table
border='1'><tr><th>ID</th><th>Username</th><th>Password</th><th>Role</th><th
>Actions</th></tr>");

        try {
            // 1. Establish Database Connection
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection conn =
DriverManager.getConnection("jdbc:mysql://localhost:3306/cse3023", "root",
"admin");

            // 2. Prepare and Execute SQL Query
            Statement stmt = conn.createStatement();
            ResultSet rs = stmt.executeQuery("SELECT * FROM users");

            // 3. Iterate through results and build table rows
            while (rs.next()) {
                int id = rs.getInt("id");
                out.println("<tr>");
                out.println("<td>" + id + "</td>");
                out.println("<td>" + rs.getString("username") + "</td>");
                out.println("<td>" + rs.getString("password") + "</td>");
```

```
            out.println("<td>" + rs.getString("roles") + "</td>");

            // 4. Add Action Links for Edit and Delete (passes ID in the URL)
            out.println("<td><a href='UpdateServlet?id=" + id + "'>Edit</a> | ");
            out.println("<a href='DeleteServlet?id=" + id + "'>Delete</a></td>");
            out.println("</tr>");
        }
        out.println("</table>");
        out.println("<br><a href='index.html'>Add New User</a>");

        // 5. Close Connections
        rs.close();
        stmt.close();
        conn.close();

    } catch (Exception e) {
        e.printStackTrace();
    }
  }
}
```

```
package com.lab.controller;

import java.io.IOException;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/**
```

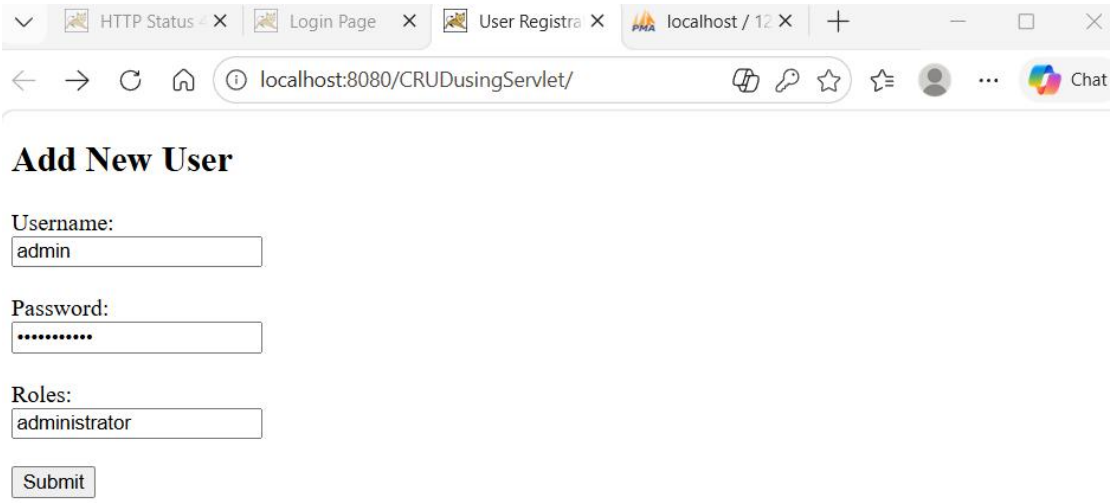
```
 *
 * @author Ikhwan
 */
@WebServlet(name = "DeleteServlet", urlPatterns = {"/DeleteServlet"})
public class DeleteServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        // 1. Get ID from the URL
        String id = request.getParameter("id");

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection conn = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/cse3023", "root", "admin");

            // 2. Execute DELETE statement
            String sql = "DELETE FROM users WHERE id=?";
            PreparedStatement pstmt = conn.prepareStatement(sql);
            pstmt.setInt(1, Integer.parseInt(id));

            pstmt.executeUpdate(); //

            pstmt.close();
            conn.close();

            // 3. Return to the list
            response.sendRedirect("ViewServlet");

        } catch (Exception e) {
            e.printStackTrace();
        }
    }


}
```

```
package com.lab.controller;

import java.io.IOException;
import java.io.PrintWriter;
import java.sql.*;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
/**
 *
 * @author Ikhwan
 */
@WebServlet(name = "UpdateServlet", urlPatterns = {"/UpdateServlet"})
public class UpdateServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
     response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String id = request.getParameter("id");

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection conn = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/cse3023", "root", "admin");

            String sql = "SELECT * FROM users WHERE id=?";
            PreparedStatement pstmt = conn.prepareStatement(sql);
            pstmt.setInt(1, Integer.parseInt(id));
            ResultSet rs = pstmt.executeQuery();

            if (rs.next()) {
                String currentRole = rs.getString("roles");
                out.println("<h2>Update User</h2>");
                out.println("<form action='UpdateServlet' method='POST'>");
                // Hidden field to keep track of which ID we are updating
                out.println("<input type='hidden' name='id' value='" + rs.getInt("id") + "'>");
                out.println("Username: <input type='text' name='username' value='" +
rs.getString("username") + "' required><br><br>");
```

```
            out.println("Password: <input type='text' name='password' value='" +
rs.getString("password") + "' required><br><br>");

            out.println("Role: <select name='roles'>");
            out.println("<option value='Admin' " + (currentRole.equals("Admin") ?
"selected" : "") + ">Admin</option>");
            out.println("<option value='Staff' " + (currentRole.equals("Staff") ?
"selected" : "") + ">Staff</option>");
            out.println("<option value='Student' " + (currentRole.equals("Student") ?
"selected" : "") + ">Student</option>");
            out.println("</select><br><br>");

            out.println("<input type='submit' value='Update User'>");
            out.println("</form>");
        }
        conn.close();
    } catch (Exception e) { e.printStackTrace(); }
  }

  @Override
  protected void doPost(HttpServletRequest request, HttpServletResponse
response)
        throws ServletException, IOException {
    String id = request.getParameter("id");
    String username = request.getParameter("username");
    String password = request.getParameter("password");
    String roles = request.getParameter("roles");

    try {
        Class.forName("com.mysql.cj.jdbc.Driver");
        Connection conn = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/cse3023", "root", "admin");

        String sql = "UPDATE users SET username=?, password=?, roles=? WHERE
id=?";
        PreparedStatement pstmt = conn.prepareStatement(sql);
        pstmt.setString(1, username);
        pstmt.setString(2, password);
        pstmt.setString(3, roles);
        pstmt.setInt(4, Integer.parseInt(id));
```

```
            pstmt.executeUpdate();
            conn.close();
            response.sendRedirect("ViewServlet");
        } catch (Exception e) { e.printStackTrace(); }
    }
}
```

OUTPUT:



localhost:8080/CRUDusingServlet/UpdateServlet?id=1

Update User

Username: admin

Password: password123

Role: Admin

Update User

localhost:8080/CRUDusingServlet/ViewServlet

User List

ID	Username	Password	Role	Actions
2	admin	password123	administrator	Edit | Delete

Add New User

localhost:8080/CRUDusingServlet/View... Summarize

User List

ID	Username	Password	Role	Actions
2	admin	password123	administrator	Edit | Delete
3	Ikhwan	Ikhwan2005@	administrator	Edit | Delete

Add New User

TASK5: Using Servlets, DAO and Java Beans for Database CRUD Operations

CODE:

```
<!DOCTYPE html>
<html>
   <head>
      <title>User Registration</title>
      <meta charset="UTF-8">
   </head>
   <body>
      <h2>Add New User</h2>
      <form action="InsertServlet" method="POST">
         <label>Username:</label><br>
         <input type="text" name="username" required><br><br>

         <label>Password:</label><br>
         <input type="password" name="password" required><br><br>

         <label>Roles:</label><br>
         <input type="text" name="roles" required><br><br>

         <input type="submit" value="Submit">
      </form>
   </body>
</html>


package com.lab.dao;

import com.lab.model.User;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.util.ArrayList;
import java.util.List;
/**
 *
 * @author Ikhwan
 */
public class UserDAO {
   private String jdbcURL = "jdbc:mysql://localhost:3306/CSE3023";
   private String jdbcUsername = "root";
   private String jdbcPassword = "admin";

   // Method untuk mendapatkan sambungan Database
   protected Connection getConnection() {
   Connection connection = null;
   try {
      Class.forName("com.mysql.cj.jdbc.Driver");
```

```
        connection = DriverManager.getConnection(jdbcURL,
        jdbcUsername, jdbcPassword);
        } catch (SQLException | ClassNotFoundException e) {
        e.printStackTrace();
        }
        return connection;
}

    // CREATE: Masukkan pengguna baharu
    public void insertUser(User user) {
    String sql = "INSERT INTO users (username, password, roles) VALUES (?, ?, ?)";
    try (Connection conn = getConnection();PreparedStatement pstmt = conn.prepareStatement(sql)) {
    pstmt.setString(1, user.getUsername());
    pstmt.setString(2, user.getPassword());
    pstmt.setString(3, user.getRoles());

    pstmt.executeUpdate();

    } catch (SQLException e) {
    e.printStackTrace();
    }
}

    // READ: Ambil satu pengguna berdasarkan ID (Untuk Update)
    public User selectUser(int id) {
    User user = null;
    String sql = "SELECT id, username, password, roles FROM users WHERE id =?";
    try (Connection conn = getConnection();
        PreparedStatement pstmt = conn.prepareStatement(sql)) {

    pstmt.setInt(1, id);
    ResultSet rs = pstmt.executeQuery();

    if (rs.next()) {
        String username = rs.getString("username");
        String password = rs.getString("password");
        String roles = rs.getString("roles");
        user = new User(id, username, password, roles);
        }
    } catch (SQLException e) {
    e.printStackTrace();
    }
    return user;
    }
    // READ: Ambil senarai semua pengguna (Untuk View)
    public List<User> selectAllUsers() {
        List<User> users = new ArrayList<>();
        String sql = "SELECT * FROM users";
        try (Connection conn = getConnection();PreparedStatement pstmt = conn.prepareStatement(sql)) {

        ResultSet rs = pstmt.executeQuery();

        while (rs.next()) {
            int id = rs.getInt("id");
            String username = rs.getString("username");
            String password = rs.getString("password");
            String roles = rs.getString("roles");
            users.add(new User(id, username, password, roles));
        }
    } catch (SQLException e) {
```

```
        e.printStackTrace();
    }
    return users;
}

    // UPDATE: Kemaskini maklumat pengguna
    public boolean updateUser(User user) {
    boolean rowUpdated = false;
    String sql = "UPDATE users SET username = ?, password = ?,roles = ? WHERE id = ?";
    try (Connection conn = getConnection();PreparedStatement pstmt = conn.prepareStatement(sql)) {

    pstmt.setString(1, user.getUsername());
    pstmt.setString(2, user.getPassword());
    pstmt.setString(3, user.getRoles());
    pstmt.setInt(4, user.getId());

    rowUpdated = pstmt.executeUpdate() > 0;
    } catch (SQLException e) {
        e.printStackTrace();
    }
    return rowUpdated;
    }
    // DELETE: Padam pengguna
    public boolean deleteUser(int id) {
        boolean rowDeleted = false;
        String sql = "DELETE FROM users WHERE id = ?";
        try (Connection conn = getConnection();PreparedStatement pstmt = conn.prepareStatement(sql)) {

            pstmt.setInt(1, id);
            rowDeleted = pstmt.executeUpdate() > 0;
            } catch (SQLException e) {
                e.printStackTrace();
            }
            return rowDeleted;
    }
}


package com.lab.model;

/**
 *
 * @author Ikhwan
 */
public class User {
    private int id;
    private String username;
    private String password;
    private String roles;

    // Constructor Kosong (Wajib untuk JavaBean)
   public User() {
    }

    // Constructor dengan ID (Untuk Update & Delete)
    public User(int id, String username, String password, String roles) {
        this.id = id;
        this.username = username;
        this.password = password;
        this.roles = roles;
```

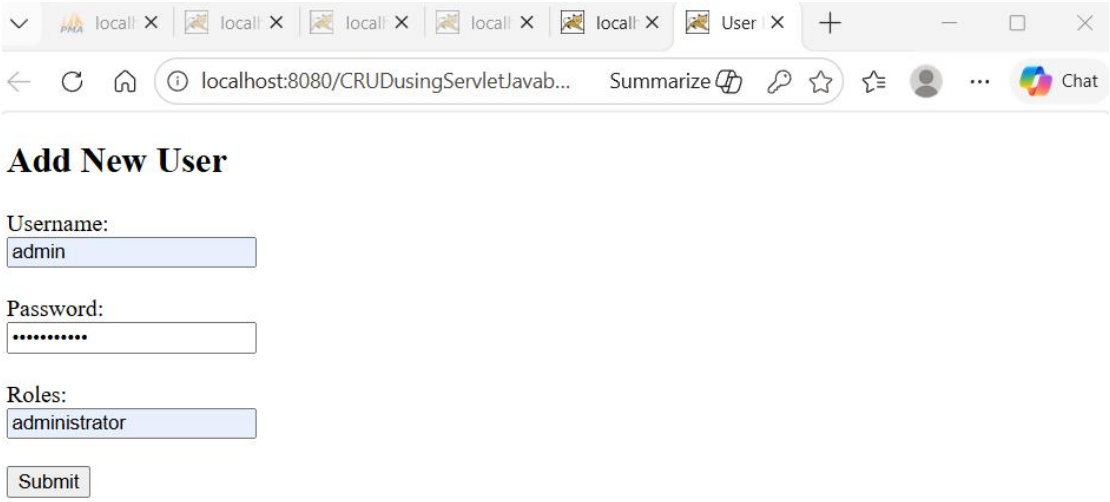
```
    }

    // Constructor tanpa ID (Untuk Insert)
    public User(String username, String password, String roles) {
    this.username = username;
    this.password = password;
    this.roles = roles;
    }

    // Getter dan Setter
    public int getId() {
    return id;
    }

    public void setId(int id) {
    this.id = id;
    }
    public String getUsername() {
    return username;
    }

    public void setUsername(String username) {
    this.username = username;
    }

    public String getPassword() {
    return password;
    }

    public void setPassword(String password) {
    this.password = password;
    }

    public String getRoles() {
    return roles;
    }

    public void setRoles(String roles) {
    this.roles = roles;
    }
}

package com.lab.controller;

import com.lab.dao.UserDAO;
import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet(name = "DeleteServlet", urlPatterns = {"/DeleteServlet"})
public class DeleteServlet extends HttpServlet {
    private UserDAO userDAO;

    public void init() {
        userDAO = new UserDAO();
    }
```

```
    protected void doGet(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {

        int id = Integer.parseInt(request.getParameter("id"));
        userDAO.deleteUser(id); // Panggil DAO

        response.sendRedirect("ViewServlet");
    }
}


package com.lab.controller;

import com.lab.dao.UserDAO;
import com.lab.model.User;
import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet(name = "InsertServlet", urlPatterns = {"/InsertServlet"})
public class InsertServlet extends HttpServlet {

    private UserDAO userDAO;

    public void init() {
        userDAO = new UserDAO();
    }

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        String username = request.getParameter("username");
        String password = request.getParameter("password");
        String roles = request.getParameter("roles");

        User newUser = new User(username, password, roles);
        userDAO.insertUser(newUser); // Panggil DAO, bukan JDBC

        response.sendRedirect("ViewServlet");
    }
}


package com.lab.controller;

import com.lab.dao.UserDAO;
import com.lab.model.User;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet(name = "UpdateServlet", urlPatterns = {"/UpdateServlet"})
public class UpdateServlet extends HttpServlet {
```

```
    private UserDAO userDAO;

    public void init() {
        userDAO = new UserDAO();
    }

    // Paparkan borang dengan data sedia ada
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        int id = Integer.parseInt(request.getParameter("id"));

        User existingUser = userDAO.selectUser(id); // Panggil DAO

        if (existingUser != null) {
            out.println("<h2>Update User (Using DAO)</h2>");
            out.println("<form action='UpdateServlet' method='POST'>");
            out.println("<input type='hidden' name='id' value='" + existingUser.getId() + "'>");
            out.println("Username: <input type='text' name='username' value='"
                    + existingUser.getUsername() + "' required><br><br>");
            out.println("Password: <input type='text' name='password' value='"
                    + existingUser.getPassword() + "' required><br><br>");

            String currentRole = existingUser.getRoles();
            out.println("Role: <select name='roles'>");
            out.println("<option value='Admin' " + (currentRole.equals("Admin") ? "selected" : "") +
">Admin</option>");
            out.println("<option value='Staff' " + (currentRole.equals("Staff") ? "selected" : "") +
">Staff</option>");
            out.println("<option value='Student' " + (currentRole.equals("Student") ? "selected" : "") +
">Student</option>");
            out.println("</select><br><br>");

            out.println("<input type='submit' value='Update User'>");
            out.println("</form>");
        }
    }

    // Simpan data yang telah dikemaskini
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        int id = Integer.parseInt(request.getParameter("id"));
        String username = request.getParameter("username");
        String password = request.getParameter("password");
        String roles = request.getParameter("roles");

        User user = new User(id, username, password, roles);
        userDAO.updateUser(user); // Panggil DAO

        response.sendRedirect("ViewServlet");
    }
}

package com.lab.controller;

import com.lab.dao.UserDAO;
import com.lab.model.User;
```

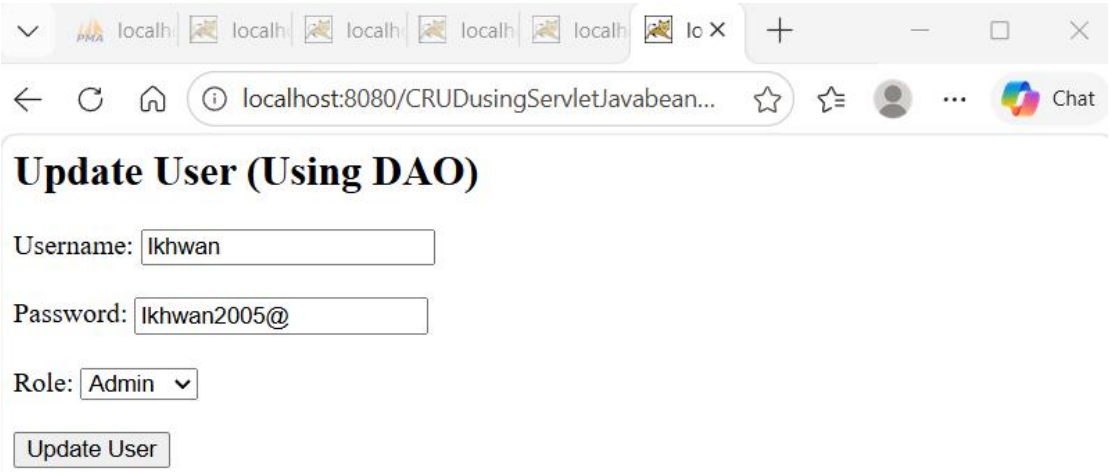
```
import java.io.IOException;
import java.io.PrintWriter;
import java.util.List;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet(name = "ViewServlet", urlPatterns = {"/ViewServlet"})
public class ViewServlet extends HttpServlet {
    private UserDAO userDAO;

    public void init() {
        userDAO = new UserDAO();
    }

    protected void doGet(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        List<User> listUser = userDAO.selectAllUsers(); // Panggil DAO

        out.println("<h2>User List (Using DAO)</h2>");
        out.println("<table border='1'><tr><th>ID</th><th>Username</th>"
                + "<th>Password</th><th>Role</th><th>Actions</th></tr>");

        for (User user : listUser) {
            out.println("<tr>");
            out.println("<td>" + user.getId() + "</td>");
            out.println("<td>" + user.getUsername() + "</td>");
            out.println("<td>" + user.getPassword() + "</td>");
            out.println("<td>" + user.getRoles() + "</td>");
            out.println("<td><a href='UpdateServlet?id=" + user.getId()
                    + "'>Edit</a> | ");
            out.println("<a href='DeleteServlet?id=" + user.getId() + "'>"
                    + "Delete</a></td>");
            out.println("</tr>");
        }
        out.println("</table>");
        out.println("<br><a href='index.html'>Add New User</a>");
    }
}
```

OUTPUT:


localhost:8080/CRUDusingServletJavab...
Summarize
Chat
Add New User
Username:
admin
Password:
Roles:
administrator
Submit

localhost:8080/CRUDusingServl...
Chat
User List (Using DAO)
ID Username Password Role Actions
2 admin password123 administrator Edit | Delete
3 Ikhwan Ikhwan2005@ administrator Edit | Delete
4 admin password123 administrator Edit | Delete
Add New User

localhost:8080/CRUDusingServletJavabean...
Chat
User List (Using DAO)
ID Username Password Role Actions
3 Ikhwan Ikhwan2005@ administrator Edit | Delete
4 admin password123 administrator Edit | Delete
Add New User


localhost:8080/CRUDusingServletJavabean...
Chat
Update User (Using DAO)
Username: Ikhwan
Password: Ikhwan2005@
Role: Admin
Update User

localhost:8080/CRUDusingServl...
Chat
User List (Using DAO)
ID Username Password Role Actions
3 Ikhwan Ikhwan2005@ administrator Edit | Delete
4 admin password123 administrator Edit | Delete
6 DiniHensem yoshindo administrator Edit | Delete
Add New User

EXERCISE:

CODE:

```
<!DOCTYPE html>
<html>
    <head><title>Inventory System</title></head>
    <body>
        <h2>Product Inventory Management</h2>
        <form action="InsertProductServlet" method="POST">
            Name: <input type="text" name="name" required><br><br>
            Category: <input type="text" name="category" required><br><br>
            Price: <input type="number" step="0.01" name="price" required><br><br>
            Quantity: <input type="number" name="quantity" required><br><br>
            <input type="submit" value="Add Product">
        </form>
        <br><a href="ViewProductServlet">View All Products</a>
    </body>
</html>

package com.lab.dao;

import com.lab.model.Product;
import java.sql.*;
import java.util.ArrayList;
import java.util.List;

public class ProductDAO {
    private String jdbcURL = "jdbc:mysql://localhost:3306/cse3023";
    private String jdbcUsername = "root";
    private String jdbcPassword = "admin";

    protected Connection getConnection() {
        Connection connection = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
```

```
            connection = DriverManager.getConnection(jdbcURL, jdbcUsername,
jdbcPassword);
        } catch (SQLException | ClassNotFoundException e) {
            e.printStackTrace();
        }
        return connection;
    }

    // CREATE: Insert a new product
    public void insertProduct(Product product) {
        String sql = "INSERT INTO products (name, category, price, quantity) VALUES
(?, ?, ?, ?)";
        try (Connection conn = getConnection();
             PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setString(1, product.getName());
            pstmt.setString(2, product.getCategory());
            pstmt.setDouble(3, product.getPrice());
            pstmt.setInt(4, product.getQuantity());
            pstmt.executeUpdate();
        } catch (SQLException e) {
            e.printStackTrace();
        }
    }

    // READ: Select a single product (Used for Update Form)
    public Product selectProduct(int id) {
        Product product = null;
        String sql = "SELECT * FROM products WHERE id = ?";
        try (Connection conn = getConnection();
             PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setInt(1, id);
            ResultSet rs = pstmt.executeQuery();
            if (rs.next()) {
                product = new Product(
                    rs.getInt("id"),
                    rs.getString("name"),
                    rs.getString("category"),
                    rs.getDouble("price"),
                    rs.getInt("quantity")
                );
```

```
            }
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return product;
    }

    // READ: Select all products
    public List<Product> selectAllProducts() {
        List<Product> products = new ArrayList<>();
        String sql = "SELECT * FROM products";
        try (Connection conn = getConnection();
             Statement stmt = conn.createStatement();
             ResultSet rs = stmt.executeQuery(sql)) {
            while (rs.next()) {
                products.add(new Product(
                    rs.getInt("id"),
                    rs.getString("name"),
                    rs.getString("category"),
                    rs.getDouble("price"),
                    rs.getInt("quantity")
                ));
            }
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return products;
    }

    // UPDATE: Update product details
    public boolean updateProduct(Product product) {
        boolean rowUpdated = false;
        String sql = "UPDATE products SET name=?, category=?, price=?, quantity=?
WHERE id=?";
        try (Connection conn = getConnection();
             PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setString(1, product.getName());
            pstmt.setString(2, product.getCategory());
            pstmt.setDouble(3, product.getPrice());
            pstmt.setInt(4, product.getQuantity());
```

```
            pstmt.setInt(5, product.getId());
            rowUpdated = pstmt.executeUpdate() > 0;
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return rowUpdated;
    }

    // DELETE: Delete product
    public boolean deleteProduct(int id) {
        boolean rowDeleted = false;
        String sql = "DELETE FROM products WHERE id=?";
        try (Connection conn = getConnection();
             PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setInt(1, id);
            rowDeleted = pstmt.executeUpdate() > 0;
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return rowDeleted;
    }
}

package com.lab.model;

public class Product {
    private int id;
    private String name;
    private String category;
    private double price;
    private int quantity;

    public Product() {}

    public Product(int id, String name, String category, double price, int quantity) {
        this.id = id;
        this.name = name;
        this.category = category;
        this.price = price;
        this.quantity = quantity;
```

```
    }

    public Product(String name, String category, double price, int quantity) {
        this.name = name;
        this.category = category;
        this.price = price;
        this.quantity = quantity;
    }

    public int getId() { return id; }
    public void setId(int id) { this.id = id; }
    public String getName() { return name; }
    public void setName(String name) { this.name = name; }
    public String getCategory() { return category; }
    public void setCategory(String category) { this.category = category; }
    public double getPrice() { return price; }
    public void setPrice(double price) { this.price = price; }
    public int getQuantity() { return quantity; }
    public void setQuantity(int quantity) { this.quantity = quantity; }
}

package com.lab.controller;

import com.lab.dao.ProductDAO;
import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/DeleteProductServlet")
public class DeleteProductServlet extends HttpServlet {
    private ProductDAO productDAO;

    public void init() {
        productDAO = new ProductDAO();
    }

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
```

```
        throws ServletException, IOException {

    int id = Integer.parseInt(request.getParameter("id"));

    // Call DAO to delete
    productDAO.deleteProduct(id);

    // Refresh the list
    response.sendRedirect("ViewProductServlet");
  }
}

package com.lab.controller;

import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/InsertProductServlet")
public class InsertProductServlet extends HttpServlet {
  private ProductDAO productDAO;

  public void init() {
    productDAO = new ProductDAO();
  }

  protected void doPost(HttpServletRequest request, HttpServletResponse
response)
        throws ServletException, IOException {

    // Retrieve parameters from form
    String name = request.getParameter("name");
    String category = request.getParameter("category");
    double price = Double.parseDouble(request.getParameter("price"));
    int quantity = Integer.parseInt(request.getParameter("quantity"));
```

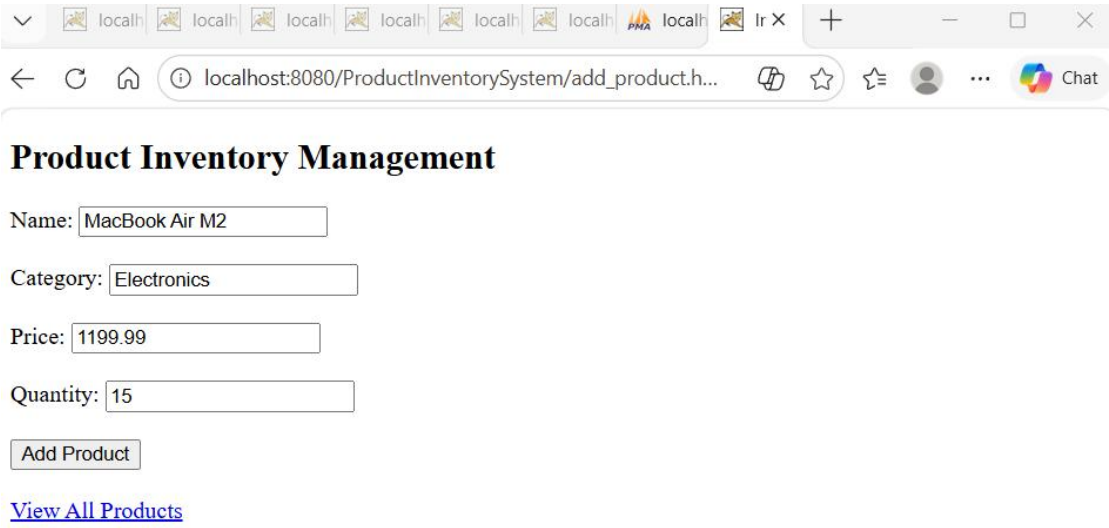
```
        // Wrap data in JavaBean
        Product newProduct = new Product(name, category, price, quantity);

        // Call DAO to save
        productDAO.insertProduct(newProduct);

        // Redirect to the list view
        response.sendRedirect("ViewProductServlet");
    }
}

package com.lab.controller;

import com.lab.dao.ProductDAO;
import com.lab.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/UpdateProductServlet")
public class UpdateProductServlet extends HttpServlet {
    private ProductDAO productDAO;

    public void init() {
        productDAO = new ProductDAO();
    }

    // Step A: Show the Edit Form
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        int id = Integer.parseInt(request.getParameter("id"));
        Product existingProduct = productDAO.selectProduct(id);

        response.setContentType("text/html");
```

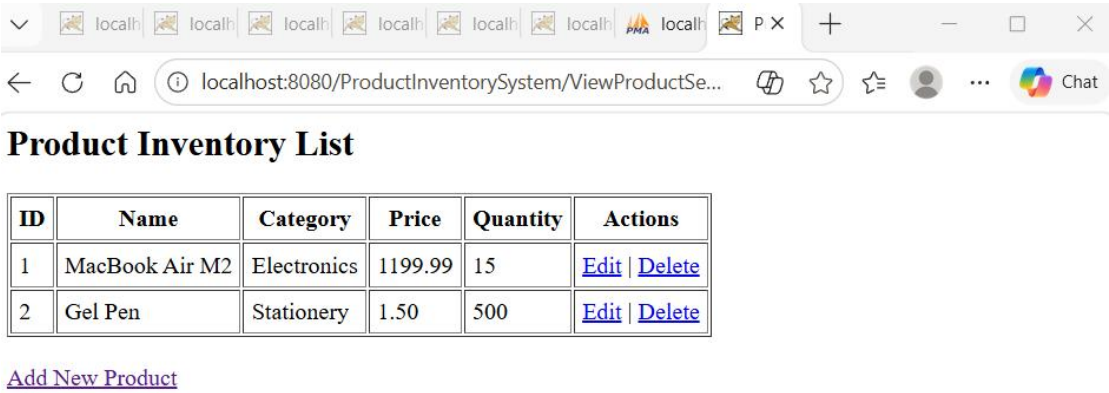
```
        PrintWriter out = response.getWriter();

        if (existingProduct != null) {
            out.println("<h2>Edit Product</h2>");
            out.println("<form action='UpdateProductServlet' method='POST'>");
            out.println("<input type='hidden' name='id' value='" + existingProduct.getId() +
"'>");
            out.println("Name: <input type='text' name='name' value='" +
existingProduct.getName() + "'><br>");
            out.println("Category: <input type='text' name='category' value='" +
existingProduct.getCategory() + "'><br>");
            out.println("Price: <input type='text' name='price' value='" +
existingProduct.getPrice() + "'><br>");
            out.println("Quantity: <input type='text' name='quantity' value='" +
existingProduct.getQuantity() + "'><br>");
            out.println("<input type='submit' value='Update Product'>");
            out.println("</form>");
        }
    }

    // Step B: Save the changes
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {

        int id = Integer.parseInt(request.getParameter("id"));
        String name = request.getParameter("name");
        String category = request.getParameter("category");
        double price = Double.parseDouble(request.getParameter("price"));
        int quantity = Integer.parseInt(request.getParameter("quantity"));

        Product product = new Product(id, name, category, price, quantity);
        productDAO.updateProduct(product);

        response.sendRedirect("ViewProductServlet");
    }
}

package com.lab.controller;

import com.lab.dao.ProductDAO;
```

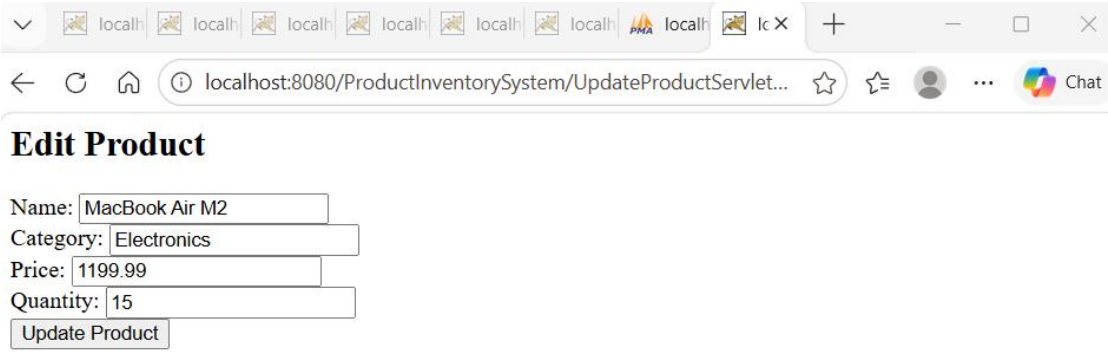
```
import com.lab.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import java.util.List;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/ViewProductServlet")
public class ViewProductServlet extends HttpServlet {
    private ProductDAO productDAO;

    @Override
    public void init() {
        productDAO = new ProductDAO();
    }

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        List<Product> listProduct = productDAO.selectAllProducts();

        out.println("<html><head><title>Product Inventory</title></head><body>");
        out.println("<h2>Product Inventory List</h2>");
        out.println("<table border='1' cellpadding='5'>");
        out.println("<tr><th>ID</th><th>Name</th><th>Category</th>"
                + "<th>Price</th><th>Quantity</th><th>Actions</th></tr>");

        for (Product product : listProduct) {
            out.println("<tr>");
            out.println("<td>" + product.getId() + "</td>");
            out.println("<td>" + product.getName() + "</td>");
            out.println("<td>" + product.getCategory() + "</td>");
            out.println("<td>" + String.format("%.2f", product.getPrice()) + "</td>");
```

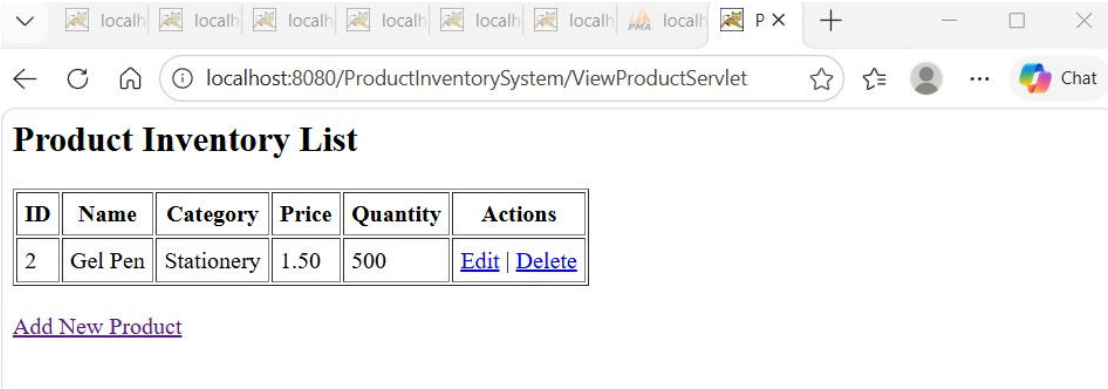
```
            out.println("<td>" + product.getQuantity() + "</td>");
            out.println("<td>");
            out.println("<a href='UpdateProductServlet?id=" + product.getId() +
"'>Edit</a> | ");
            out.println("<a href='DeleteProductServlet?id=" + product.getId() +
"'>Delete</a>");
            out.println("</td>");
            out.println("</tr>");
        }
        out.println("</table>");
        out.println("<br><a href='add_product.html'>Add New Product</a>");
        out.println("</body></html>");
    }
}
```

OUTPUT:






localhost:8080/ProductInventorySystem/UpdateProductServlet...
Chat
Edit Product
Name: MacBook Air M2
Category: Electronics
Price: 1199.99
Quantity: 15
Update Product


localhost:8080/ProductInventorySystem/ViewProductServlet
Chat
Product Inventory List
ID | Name | Category | Price | Quantity | Actions
2 | Gel Pen | Stationery | 1.50 | 500 | Edit | Delete
Add New Product